Electromagnetic waves

THESE ARE WAVES THAT CARRY ENERGY THROUGH SPACE.

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Electromagnetic (EM) waves transfer energy from one place to another. There are different types but they all travel through a vacuum, such as space, at the speed of light.

Along the spectrum

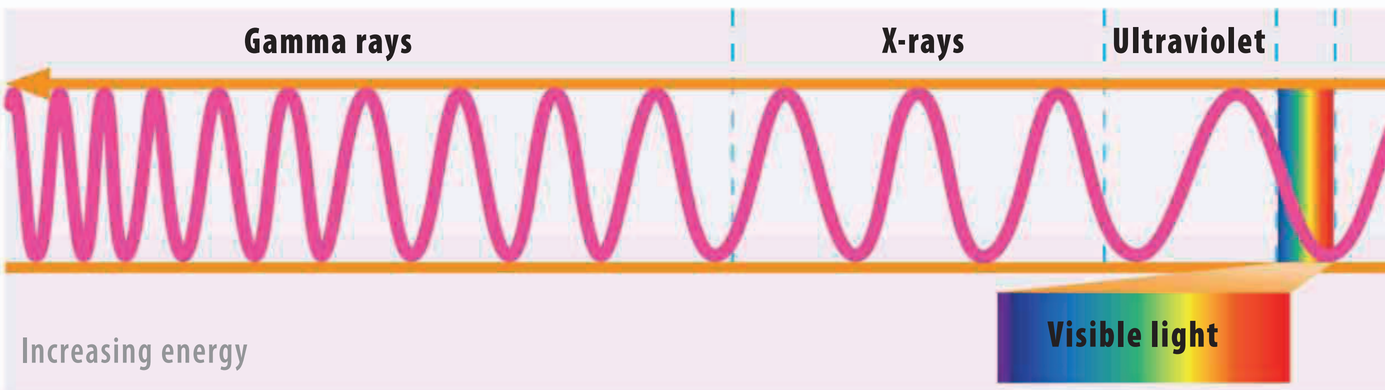
Visible light is just one type of EM wave; other types are invisible. The full range of waves, called the electromagnetic spectrum, is made up of waves of different frequencies and wavelengths. At one end are radio waves, which have the longest wavelengths and lowest frequencies. At the other end are gamma rays, with the shortest wavelengths and highest frequencies.

REAL WORLD

Snake sense

Some animals can sense infrared radiation well enough to find warm objects in the dark. Some snakes have pits —see the hollow depression on the snout of this viper—that contain heat receptor cells. At night, or when their prey is hiding, these sensors can detect the body heat of warm-blooded prey, such as mice.

▽ **Properties and uses**
The different types of electromagnetic radiation have different properties and uses, depending on their wavelength. Waves with shorter wavelengths, such as gamma rays and X-rays, can carry large amounts of energy, while longer radio waves do not.



Gamma rays
These are produced by radioactivity and can carry a lot of energy. They cannot be seen or felt but are very harmful. While they can cause cancer, they also kill cancer cells. Other uses include sterilizing food and surgical instruments.

X-rays
X-rays are used to make images of inside the body because they pass through skin and soft tissue, but are absorbed by harder materials such as bone. In high doses they can be harmful, so X-rays must be used with caution.

Ultraviolet (UV)
UV radiation is found naturally in sunlight. You cannot see it or feel it, although you can experience the effects of too much UV as sunburn. Sunblock and sunglasses should be worn to protect skin and eyes from UV damage.

Visible light
This set of wavelengths is the only one that our eyes can see. The color seen depends on the wavelength of light, with violet and blue having shorter wavelengths than green and yellow. Red has the longest wavelength of all.